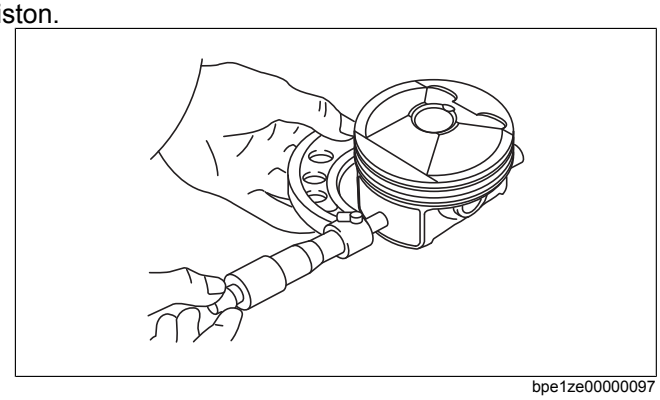
PISTON INSPECTION

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Caution
- **If the piston is replaced, replace the piston, piston pin, and the snap ring as a single component.**

1. Measure the piston outer diameter using the micrometer.
   - SKYACTIV-G 1.3: The measurement position is **11.0 mm {0.433 in}** from the lower end of the piston (area with no coating on the piston skirt) and in the thrust direction.
   - SKYACTIV-G 1.5: The measurement position is **10.0 mm {0.394 in}** from the lower end of the piston (area with no coating on the piston skirt) and in the thrust direction.
   - SKYACTIV-G 2.0, SKYACTIV-G 2.5: The measurement position is **8.0 mm {0.31 in}** from the lower end of the piston (area with no coating on the piston skirt) and in the thrust direction.
     — If it is not within the specification, replace the piston.

Standard piston outer diameter
SKYACTIV-G 1.3: 70.967—70.997 mm {2.7940—2.7951 in}
SKYACTIV-G 1.5: 74.467—74.497 mm {2.9318—2.9329 in}
SKYACTIV-G 2.0: 83.465—83.495 mm {3.2861—3.2872 in}
SKYACTIV-G 2.5: 88.965—88.995 mm {3.5026—3.5037 in}



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2. Measure the cylinder bore diameter. (See CYLINDER BLOCK INSPECTION.)
3. Calculate the cylinder-to-piston clearance from the cylinder bore diameter and the piston outer diameter.
   - If the clearance exceeds the maximum specification, replace the piston or cylinder block.

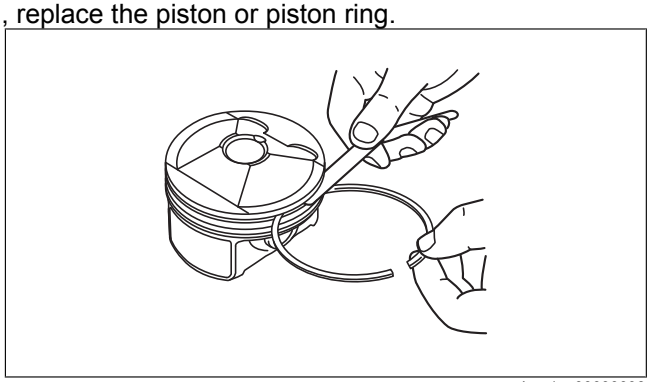
Standard clearance between piston and cylinder
SKYACTIV-G 1.3, SKYACTIV-G 1.5: 0.0230—0.0430 mm {0.00091—0.00169 in}
SKYACTIV-G 2.0, SKYACTIV-G 2.5: 0.0250—0.0450 mm {0.00099—0.00177 in}

Maximum clearance between piston and cylinder
SKYACTIV-G 1.3: 0.061 mm {0.0024 in}
SKYACTIV-G 1.5: 0.062 mm {0.0024 in}
SKYACTIV-G 2.0: 0.063 mm {0.0025 in}
SKYACTIV-G 2.5: 0.066 mm {0.0026 in}

4. Measure the piston-to-ring groove clearance along the perimeter using a feeler gauge. For the O-ring, measure the clearance with the O-ring assembled to the piston.
   - If the clearance exceeds the maximum specification, replace the piston or piston ring.

Standard clearance between piston ring and ring groove [SKYACTIV-G 1.3]
Top: 0.03—0.08 mm {0.0012—0.0031 in}
Second: 0.03—0.07 mm {0.0012—0.0027 in}
Oil: 0.04—0.12 mm {0.0016—0.0047 in}

Standard clearance between piston ring and ring groove [SKYACTIV-G 1.5, SKYACTIV-G 2.0, SKYACTIV-G 2.5]
Top: 0.04—0.08 mm {0.0016—0.0031 in}
Second: 0.03—0.07 mm {0.0012—0.0027 in}
Oil: 0.04—0.12 mm {0.0016—0.0047 in}



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Maximum clearance between piston ring and ring groove [SKYACTIV-G 1.3]
Top: 0.12 mm {0.0047 in}
Second: 0.08 mm {0.0031 in}
Oil: 0.17 mm {0.0067 in}

Maximum clearance between piston ring and ring groove [SKYACTIV-G 1.5, SKYACTIV-G 2.0, SKYACTIV-G 2.5]
Top: 0.12 mm {0.0047 in}
Second: 0.10 mm {0.0039 in}

Oil: 0.17 mm {0.0067 in}